

Response to reviewer's comments

- We would like to thank the reviewers and the Editor for the very helpful and insightful comments. The suggestions and revisions are discussed in the point-by-point reply. We also changed the title to avoid confusion with top-down control from predation and added a new schematic figure of the size-spectrum model.

Editor's Comments to the Author:

I am sorry for the delay in getting the review back. It has been a challenge to find reviewers. I have received comments from two reviewers and they are generally support of the manuscript. The reviewers made some useful suggestions to help improve the manuscript. Specifically, although the modelling approach and the scenario included were considered by the reviewer to be simple, it could be an advantage which the study may currently under-utilized and the reviewer made some suggestions on how this could be done. I think this would help improve the usefulness of the manuscript.

- Thank you we feel the changes suggested do help with the clarity and appreciate the reviewers' thoughtfulness in trying to enhance this along with maximizing potential uptake of the paper.

Reviewer 1 Comments to Author:

Lindmark et al., with their study, examine the direct and indirect effect of temperature on the physiology, populations and yields of three important fish species in the Baltic Sea. To do so, they combined empirical observations with a size spectrum model and ran model simulations with RCP8.5 regional temperature projections until 2050. The study provides novel insights on how fish size, populations and yield are affected by both direct effects of temperature on fish physiology and indirect effects via changes on plankton resources. This is a timely topic for peers in oceanography, fish ecology, fisheries and theoretical ecology. The study is very well designed and fits with the aim and scope of the journal.

The manuscript is well written. The abstract is a great summary of the work. The authors linked their study wonderfully to state of the art in the introduction and discussion. The discussion also includes directions for future work. The authors provide a detailed description of their method in the methods section and supporting information. The GitHub repository has good documentation, though I didn't run the code to see if the documentation provides sufficient information for new users to run and understand the model. Overall, I recommend minor corrections, which, in my opinion, will improve readability. Most of them are related to the methods section.

I think that the length of the methods section is fine, but I often lost the aim of the study while reading the methods. I strongly recommend starting the methods with an overview paragraph following the structure of the methods section (i.e., model, temperature implementation, model calibration, model data analysis).

A summary figure of the study design will advance the manuscript and help the reader to follow the methods.

- We have added a schematic figure (Fig. 1) of the size-spectrum model and the individual-level temperature dependent bioenergetics terms. We have also added an

opening sentence to the methods, and more clearly presented our two approaches to evaluate the effects of warming. See lines 148 and 487-505.

Lines 260- 261: Change the brackets to sentence to improve readability. In mizer, individual metabolism represents...

- We have removed open bracket and edited the sentences. (Line 314-316)

“nine species × scenario combinations” (lines 421 and 428). Please refer to where the reader can find the set-up of those combinations in the methods.

- We edited this sentence to be more specific (lines 593-598)
- We also added two panels to Fig. 6 (previously Fig. 5) that show the slopes of F_{msy} and yield over temperature to better visualize the trends (easier than looking at isoclines). We also updated the figure legend text in Fig. 5 (previously Fig. 4). See also changes in the presentation of the temperature scenarios.

line 427: remove “it” after results

- This was a typo, changed to “in”.

Lines 206- 209: The sentence does not make sense. Please consider to re-write. Also, the lines 208-209 can be a second sentence.

- We have changed these lines to:
- “Note that in contrast to other MSSMs (e.g., Blanchard *et al.* 2014) species in our model do not have specific preferences for other size-structured species (all values in the species interaction matrix $\theta_{i,R}$ are set to 1). However, they have different preferences for the two background resources. This helps to account for different feeding of sprat, herring and cod on benthic and pelagic resources.” (Lines 231-234)

Lines 313- 323: Is hard to follow. It will be helpful to split the sentences.

- Sentences have been split and rearranged to make the message clearer. (Lines 388-406)

Lines 310- 312. Not sure I understood this sentence. How do you support this assumption? Metabolism can affect the mortality rates, but mortality is also a result of parasites, viruses, exposure to predation. Also, remove “in” before the ref and include the sentence in the brackets to the main sentence (with the assumption that search)

- We agree that defining a single activation energy for a rate such as mortality that is made up of very different processes is a simplification. We clarified that this temperature dependence is for background mortality only (predation mortality becomes temperature dependent via the scaling of predator feeding rates).
- The sentence was edited as follows:
“For search volume we assumed that it scales identically to maximum consumption, because both rates are related to feeding processes. For background mortality we assumed identical scaling to metabolism, as longevity is linked to life span and metabolic rate (Brown *et al.* 2004; McCoy & Gillooly 2008; Munch & Salinas 2009)”.
(Lines 392-396)

Lines 349-350: what is the purpose of this algorithm, and why did you use it? Maybe this is a common algorithm for R users but for non-users, this sentence doesn’t provide any information.

- The purpose is to find the optimal values of the parameter R_{max} , i.e., those that minimized the residual sum of squares between spawning stock biomass from the stock assessments and the model. This information about the R-function and the optimization

algorithm (which is an argument in the function) is simply for reference, since mizer (which we have forked) is an R package. We felt it was important to include for reproducibility. We have changed the sentence to more clearly state the purpose of this text. (Line 455).

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Line 459: remind the reader what the acronym TRS stands for.

- Done

Line 465: I do not understand what the meaning of the phrase in the brackets is. Do you mean that the size decline is stronger in young fish?

- Yes, that is correct. We change from “are” to “were” to match the rest of the sentence.

Line 486: add “other studies” after in and ref

- Done

Overall, the manuscript has many nominalizations (in the format of -ing) and long sentences that do not help readability. I encourage the authors to try to remove some nominalizations and split sentences. Below, I provide some examples (I don’t have to follow the proposed changes, I use them as examples of how the sentences can change).

- We appreciate all these suggestions to make the language clearer and have implemented almost all as they are.

Lines 195- 198: Change to: Satiation determines the feeding level and is represented in the model with a Holling functional response type II. Satiation varies from 0 (no satiation) to 1 (full satiation).

- Done

Lines 219- 220: End the sentence in the “search volume”. Start a new sentence for line 220 (The allometric ...).

- Done

Lines 423- 426: Change to: Cod in the scenario with only physiological scaling is the exemption. The model projects higher yields as temperature increases, due to the increase in growth rate, average size, and relative abundance of large individuals (Figs. 1-3).

- Done

Lines 300- 302: Move the “to simplify the analyses” at the beginning of the sentence.

- Done

Lines 307- 310: change to: Activation energies are estimated with uncertainty as they vary substantially between processes, species, and taxonomic groups. Here we parameterized ...

- Done, but wrote “... with uncertainty **and** they vary... (line 387)

Lines 384- 387: change to: Warming positively affects size-at-age when temperature dependence is included on metabolism, maximum consumption, search volume and mortality, with or without temperature-dependent background resource dynamics.

- Done

Lines 439-450: change to: To do so, we contrasted the effects of warming on resource productivity and individual level physiology (metabolism, feeding and background mortality)

of fish. We found that warming leads to increased size-at-age of fishes when temperature-dependence is included in physiological rates. This effect is strongest in juveniles of all three fish species. Though, despite increased growth rates, in most cases, warming leads to smaller mean body size in the population, lower spawning stock biomass (biomass of mature fish) and reduced yields.

- [Done](#)

Lines 447- 450: When temperature affects only the background resource species, the size-at-age declines for fish of all sizes.

- [Done](#)

Line 451: change to: Mechanistic models that explore

- [Done](#)

Line 523-525: We approached this by applying random parameterization, rather than fixed values of temperature dependence. To capture the uncertainty of our approach, we sampled parameters from distributions based on estimates of activation energies of physiological rates in the literature (Lindmark et al. 2021).

- [Done](#)

Line 535: change to: other similar studies that use size spectrum...

- [Done](#)

Reviewer 2 Comments to Author:

Comments to the Author

The study used a mechanistic size-based food web model to explore the warming effects on fish physiology and resource productivity and the consequences on the size structure and fisheries yield. This case study focused on the Baltic Sea provides interesting insights to better understand the physiological processes leading to changes in ecosystems size structure. The analyses appear to be competently performed, the text is well written and the results are clearly presented. However, the authors should take more advantage of the simplicity of the model to better disentangle the physiological processes at play (see general comments A). Moreover, the method section would benefit from a through revision.

General Comments

A. The study used a very simplified representation of the Baltic Sea. The simplicity allowed the authors to disentangle the temperature effects on background resources and physiological processes by testing three scenarios. However, the authors should take more advantage of the simplicity of the model structure to better understand the temperature effects on each physiological process (individual metabolism, maximum consumption, search volume and background mortality). Moreover, only one warming scenario (RCP8.5) is considered. An alternative contrasted warming scenario should be valuable to explore the effects of the magnitude of warming.

- [We have now added extra sensitivity analysis in the SI \(see Fig. S15\), where we explore the effects of each physiological process in isolation on size-at-age, with and without temperature effects on the background resources.](#)
- [However, we have not included another warming scenario besides the RCP8.5. This is because we explore the effect thoroughly in our non-time varying temperature scenarios](#)

that we use to quantify changes in yield and the maximum sustainable yield. Specifically, the x-axis in Fig. 6 (previously Fig. 5) shows temperature as a proportion of T_{ref} . It ranges from 0.75 to 1.25 in steps of 0.01, meaning we essentially explore 51 unique temperatures for each value of F (y-axis in the figure). Hence, we believe using different warming scenarios would add limited information but even more complexity in the methods and results. We therefore prefer to use RCP 8.5 in the scenario where we project forward with time-varying temperatures and many temperatures where we explore temperature effects on yields specifically.

- We clarified the number of temperature scenarios we run in the following edit:
- “We explored scenarios where temperature ranged between 0.75 to 1.25 of T_{ref} , and F_{MSY} ranged between 0.1 and 2 of $F_{MSY,T_{ref}}$. With the full factorial combination of these scenarios, this gave a total of 1989 scenarios. These simulations were done to explore the effect of temperature on fisheries yield and F_{MSY} ”. (Lines 526-538)
- We also now clarified our two different ways to expose the food web to warming in the opening of the section “*Analysis of responses to warming*”. (Lines 487-505)

B. The conceptual basis and the principles of multi-species size spectrum models (MSSM) are not well explained and described. Furthermore, there are too many equations and most of them are already described in Scott et al. 2014. The authors should focus on the core equations (eq1, eq 17...), the equations adapted specifically for this study, and the equations in which the temperature effects are incorporated. The others equations could be moved to the SI.

- We agree that most of the equations have already been introduced in Scott et al., (2014) (and references therein) in some form. However, most of them have in fact been adapted in this study from Scott et al., (2014). For instance, we have starvation mortality (Eq. 3), two background resources and a preference parameter for them (which affects the predation mortality, Eq. 5, Eq. 7) and the parameter m in Eq. 12 differs from the default value in the original mizer paper. Moreover, Eqns. 2, 6, 9, 11 and 16 have parameters that are temperature-dependent and are therefore important to have in the main text. Eq. 4 shows how fishing mortality is modelled, which is also a key equation.
- Hence, if we are to keep equations suggested by the reviewer, it would still be quite many, and then the difference between that and keeping them all (which also leads to better flow) is quite small.
- Secondly, the parameters we make temperature-dependence typically occur at lower levels. For example, maximum consumption rate affects the feeding level, which affects net energy gain used for growth and reproduction. At the highest level (Eq. 1, the McKendrick-von Foerster equation), we say dynamics of abundance-at-size are determined by growth and mortality. By only including the higher-level equations, it will be more difficult to follow the temperature-dependence section, because lower-level rates (not mentioned in the main text if those equations are moved to the SI) are the ones we scale with temperature.
- For these reasons, we think it is easier to follow the model and its temperature dependence if all equations are kept in the main text (which we were asked to clarify in a minor comment below).
- However, if the editor agrees, we would be willing to make changes with a little bit of further guidance on how to best address this.
- To help with understanding the model we have instead a conceptual diagram of the bioenergetics, the size spectrum model.

C. I would suggest to present explicitly the ecological and physiological hypothesis which led

to the choice of parameters affected by temperature and better highlight the links between the model parameters and temperature. The temperature dependence section in the SI could be moved to the main text as well as Table S3. This is central in this study. Moreover, the links between the warming-induced changes in the physiological processes and the resulting consequences on the growth rates (and size-at-age) are not clearly described and discussed.

- In the section “Temperature dependence” (Line 300) we have now more clearly stated why metabolism is temperature dependent, and why also mortality and feedings rates should be, and what background mortality represents (see also response to previous reviewer).
- When we added more text to this section in the main text, the text in the SI-section “Temperature dependence” became redundant and we removed it (but tables and illustrations are kept).
- We have also clarified how changes in growth (as evident by changes in size-at-age) emerge in the model via temperature dependence of metabolism and feeding rates. (Lines 311-314)

D. While the use of “Bottom up” is relevant here, the term “top-down” can be misleading for the readers. A top-down control is when the higher trophic levels control the lower trophic levels. This is not really what you did. So, I suggest not using “top-down” and modifying the title.

- We agree it can be confusing. We replaced top-down with “physiological” in the text and changed the title to *“Temperature impacts on fish physiology and resource abundance lead to faster growth but smaller fish sizes and yields under warming”*

E. I found that the study provides valuable insights regarding warming impacts on the size structure in marine ecosystems but a specific attention could be paid not to overinterpret the findings in the abstract and the discussion section.

- We changed from “show” to “suggest” in the final sentence in the abstract
- We edited the conclusion (786-797) section slightly
- However, we did not change anything in the main discussion with respect to this comment, because we did not know which sentences this comment referred to. In fact, we use phrases such as “in our model”, “In our modelling system”, “our findings suggest”, “reduced fisheries yields may be common in a warming world” to avoid making too broad conclusions.

Specific comments

Lines 83-85: The quantitative comparison and the parallels between growth of 223 assessed fish populations in specific locations (Free et al., 2019) and global projections of total biomass is not relevant (Lotze et al. 2019).

Free et al. (2019) looked at specific fish stocks (high trophic levels) while Lotze et al. (2019) looked at the total consumer biomass. Furthermore, the projections of Lotze et al. (2019) cover the global ocean while in Free et al. (2019), the authors looked some specific locations.

- When we cite Free *et al.*, (2019), we refer to the results on MSY (maximum sustainable yield), not their findings on “productivity” (which we assume is meant by growth in this comment and one further down). We agree that the many differences between the studies make quantitative comparisons difficult. However, data from Free *et al.*, (2019) are used in Lotze et al., (2019) as “empirical validation of ensemble projections”, and Lotze et al., (2019) also say:

“Recent results hindcasting temperature effects on biomass of 235 assessed fish stocks found a 4.1% drop in MSY from 1930 to 2010, a period that saw an average 0.6 °C of warming (30 [Free et al. (2019)]). This reconstructed rate of change is consistent with our 5% drop in biomass per 1 °C of warming throughout the 21st century”

Hence, we do think there is merit in comparing how these are “*matched in magnitude and direction*” (as we phrase it). However, we changed from “productivity (population growth at a given biomass)” to “the maximum sustainable yield” to better match the units (biomass).

We feel that references to findings in other empirical and modelling studies that assessed climate impacts on fish stocks and yields are important, to put a perspective on our modelling approaches.

Lines 86-87: No. if the declines in productivity and biomass were mostly induced by the changes in primary production, the magnitude of the changes in productivity and biomass would be the same as the one of the changes in primary production. This is not the case. On the contrary, Lotze et al. 2019 highlighted that the changes in biomass are amplified at higher trophic levels compared with phytoplankton and that the projected changes are induced by both the changes in primary production and temperature.

- We agree this was unclear and both drivers are important (and vary across models). We have removed “mostly” from the sentence and now write: “*Across a range of process-based ecosystem models, declines in productivity of fish biomass and abundance of large fish, have been ~~mostly~~ linked to combined changes in temperature, primary production, zooplankton abundance and temperature*”. (Line 90)
- We cite several papers (Barange et al., 2014; Blanchard et al., 2012; Lotze et al., 2019; Woodworth-Jefcoats et al., 2013, 2015).
- While also Lotze et al., (2019) highlights declines in primary production as the first mechanism (“*This general trend can be explained by warming causing increased ocean stratification, which reduces nutrient availability in the upper ocean, leading to decreased primary production and lower energy supply for higher trophic levels (14, 18), and changes in metabolic rates, among others*”), it is true there is trophic amplification due to other factors, that in turn vary among models.
- The need for improved understanding of the ways in which links to resource and temperature effects interact and contribute to model uncertainty has been highlighted in more recent FishMIP model papers (Heneghan et al., 2021; Tittensor et al., 2021 IPCC AR6). Our paper aims to systemically disentangle these different drivers and interactions, thus contributing towards improved understanding and reduction of uncertainties.

Line 113: Add more context regarding the Tu et al. study.

- Done. “Also Tu *et al.* (2018) found that temperature had a relatively minor effect on fish size structure compared to fishing, when assessing 28 stocks from the North Sea, US west coast and Eastern Bering Sea.” (Line 114)

Lines 141-146: This is results and should be moved to the results, discussion or conclusion sections (“find that not only ...”).

- We removed the final sentences from the introduction that described results. We did not move them because the message is already conveyed in the discussion and conclusion.

Line 155: Define what is a “background resource” for GCB readers who are not specialists in size-spectrum models. Also, a conceptual schema of the model may be useful.

- The background spectra are defined one sentence below: “*The pelagic background resource spectrum represents mainly phyto- and zooplankton while the benthic background resource spectrum represents benthic invertebrates, gobiidae and small flatfish*” (Line 170-172).
- We have also added similar text as a reminder in the caption of our conceptual diagram, which shows the resource spectrum visually.

Line 167-168: Is the parametrization based on another model?

- No, we use a mix of default parameters, parameters estimated specifically for this study and parameters found through the calibration (optimization procedure). The source description, unit and value for all parameters are found in Appendix Tables S1-S2.

Line 172: Please give more explanation of the “bioenergetic principles” and the underlying assumption(s)?

- We added a parenthesis: “*bioenergetic principles (explicit modelling of energy acquisition and use)*”. (Line 185)
- We also added a schematic figure of the model (individual-level bioenergetics and size-spectra)
- We are not sure if this comment referred to a specific equation that needed clarification of the assumptions or if this was a general comment.

Lines 173-257: Several terms of the equations are not cited and described in the text. Could please add them and provide some details when it is necessary.

- We have now defined $N_i(w)$ for Eq. 1.
- Eq. 3 introduces terms that are defined in Eq. 11 (as stated in the text). While this is not optimal, we accept this to introduce the mortality terms in the same place.
- Predation mortality (Eq. 5) contains many terms; hence it is inevitable that some are explained further down.
- Eq. 9 is an equation default to mizer that is used to calculate gamma. Since this equation has no bearing on other parts of the model or the study, we provide references (Andersen & Beyer, 2006; Scott et al., 2018) instead of defining and explaining it in detail.
- We added an explanation for m in Eq. 12.
- In Eq. 15 we correct a subscript to match Eq. 14.
- Besides these changes, we cannot find any terms that are not defined (but acknowledge that not all terms can be defined just after mentioned due to the relative complex model).
- Were there any specific terms that you had in mind?

Line 177: Explain what is the “background mortality”.

- We have clarified now that it represents mortality related to ageing and diseases (Line 197)

Line 180: Is the allometric constant species specific?

- Yes, it is indexed by i (species). See also Eq. 6

Line 189: Could you explain why you chose this type of selectivity?

- This was for simplicity, and because fisheries tend to target mature individuals.
- “[...] where S_i is the selectivity (for simplicity, we assumed knife-edge selectivity with weight at first catch corresponding to weight at maturation) [...]” (Line 208).

Line 192: The notation is confusing. You gave the background, starvation and fishing mortalities for the species i , and here you provide the predation mortality for the species j .

- Yes, we see how this can be confusing. However, we chose to write it from this perspective (predators i and prey j) because the predation mortality also equals the amount consumed by a predator species. This is used in equations related to net energy gain, growth, and reproduction, which are subscripted with i . If we would use i for the species exposed to predation, the similar confusion would occur in more instances when describing consumption.

Line 193: $(\Phi)_j$ in the text (line 193) and $(\Phi)_i$ in the equation (line 192).

- Thank you, this was a typo.

Line 263: Can you provide the underlying ecological hypothesis regarding the temperature effects on the different model parameters?

- We say in the opening sentence that temperature affects metabolism, which in turn affects other processes, in line with the metabolic theory of ecology (lines 301-303)

Lines 278-279: As far as I understand, the way to incorporate temperature effects on the amount of available background resources is very different. The temperature effect on the background resources is a change in resource productivity.

- Yes, the resources are affected by temperature via their growth rate and carrying capacity, not the biomass itself in our model. We edited this sentence to “Temperature also affects plankton and benthos organisms, represented in our model through background resources”, and then describe how in the paragraph. (Line 342-343)

Line 284: Net primary production is (can be) an output from earth-system models. This is the category of forcings you described in the previous sentence? If not, can you precise the differences?

- Yes, that is the same category. We have removed the sentence and move the references to appear after the “earth-system model”. (Line 347)

Line 284: Precise which “growth rates”.

- No longer in the text because of the above edit.

Line 288: It is not clear. Here you presented how you applied the warming effects on “bottom-up” not on “top down” processes.

- We agree it was not clear. It has been edited now: “*In order to isolate the effects of temperature on resource and physiological processes,*”. (Line 350-351)

Lines 297-298: It is not correct. The overall projected decline in primary production is mainly induced by the projected increase in stratification (inducing a nutrient limitation) (see Kwiatkowski et al. 2019).

- The sentence was pointing to the similar consequences for primary production, not that the mechanisms was the same. We have deleted this sentence to avoid confusion.

Lines 305-306: The comparison with these studies is not relevant at all. Also, I do not know

what is compared the carrying capacity with decrease in consumer biomass (Lotze et al., 2019) or the net primary production (Lotze et al., 2019) or the growth rate (Free et al., 2019)

- Yes, we agree, likely this citation was an error. We meant to cite only Heneghan *et al.*, (2019) for the specific comparison about the decline in the elevation of the size spectrum. We removed the reference to Lotze *et al.*, (2019) and Free *et al.*, (2019). (Line 365)

Line 363: Provide more details regarding the forcings. Did you use the mean historical fishing mortalities? What is the “centred” sea surface temperature? And it would be good to give the magnitude of warming in 2050 (delta temperature relative to the reference period)?

- We have clarified that we used annual fishing mortalities in the period 1974-2012) (Lines 506-507) and how the relative time series was scaled. Note we also changed the arbitrary T_{ref} to make it easier to describe how the scaling was done (now 10°C instead of ~10.1°C). (Line 509-511)

Line 368: It is not clear to me how you calculated F_{msy} ? Why did not you use the F_{msy} of the stock assessments? Did you compare your estimates with the F_{msy} of the stock assessments?

- We have clarified how F_{msy} (fishing mortality resulting in the highest yield) was calculated (line 481-484).
“The fishing mortality leading to the highest long-term yield (F_{MSY}) from the model (estimated for one species at the time while keeping each species at their mean assessment F_{MSY}) were in agreement with the assessment F_{MSY} for sprat and herring. For cod, F_{MSY} is lower in the size-spectrum model than in stock assessments.”
- The reason for using the size-spectrum F_{msy} rather than the stock-assessment F_{msy} is because they differ and are calculated in different ways. For instance, F_{msy} values from assessments differ depending on whether predation (a key features in size-spectrum models) is explicitly considered. Because of these differences in values, cod would be overfished, while sprat and herring could be fished below F_{msy} (depending on which assessment reference points we had chosen). This would confound the analysis of the food web in 2050.
- Yes, the estimates for F_{msy} are compared in Fig. S9.

Line 370: Specify the period over which you evaluated the warming. As I can see on the legend of the figures, you looked at the weight-at-age, population mean weight and abundance-at-weight in 2050. Overall, can you provide more details regarding what you focused on in the results and on the scenarios.

- Clarified that we evaluate the food web in 2050.
- We are unsure what is meant by the last sentence “Overall, can you provide more details regarding what you focused on in the results and on the scenarios”

Line 380: Could you provide some details in the manuscript regarding the result of the model calibration/validation? It can be a summary of the section “results from calibration procedure” in the SI.

- We have now edited that section and brought in some of the text from the SI summarizing the calibration procedure. (See e.g., lines 434-436 for the paragraph opening, 450-455 for a clarification of the optimization procedure and 467-475 for a brief overview of the calibration results).

Line 387: There is only “one scenario without temperature dependent physiological

processes” (the yellow line in fig 1).

- Correct, has been edited.

Line 408: Please provide more details regarding the temperature effects on growth rates in the M&M. You do not apply temperature effects directly on growth rates so it is not clear how temperature affects growth rates. Temperature affects the rates of individual metabolism and maximum consumption which affects the net energy which affect the growth rate. It is not explicit at all in the manuscript.

- We have now illustrated how feeding, metabolism and energy allocation determine growth rates in our new schematic figure (Fig. 1).
- We also added a sentence clarifying how metabolism and consumption are key terms in the equations for net energy gains and growth rates (in materials and methods). (Line 311-312)
- Lastly, we clarify that growth rates become temperature dependent via these rates, we do not have a direct temperature dependence of growth. (Line 312-314)

Line 438: You did not show the warming effect on growth rates in the manuscript, in the figure or in the SI. You showed the warming effects on size-at-age which is directly related to growth rates but it is not explicit in the manuscript.

- Yes, true. We have clarified on line 144 that we investigate growth-trajectories (size-at-age), and we also remind this is the case in the results (Line 543) and discussion. However, we do not omit the term growth rate completely, because larger size-at-age can only occur by faster growth rates.

Line 458: In Lefort et al., 2015, the authors used the APECOSM and in the model all physiological rates are temperature-dependent. As far as I remembered, many ecosystem models accounted and showed that warming affects metabolism with increasing metabolic rates. See Heneghan et al 2021.

- Yes it is true many size-spectrum models and ecosystem models have temperature dependence of physiological rates (Woodworth-Jefcoats et al., 2019). The key difference here is that we contrast that to scenarios with and without resource temperature dependence, incorporate uncertainty in temperature-sensitivity of rates and simulate 51 temperatures as well as time-varying temperatures to 2050.
- Although, it is unclear to us what this comment refers to. On line 454 we cited Lefort *et al.*, (2015) (now this is line 631) when we say declines in community wide body size are due to food limitation or changes in the size-structure.

Lines 472-474: Could you elaborate and provide reference?

- With “general” we referred to our projections with random activation energies, not as a general result. We have clarified that this was a result of ours.

Line 480 : “Higher” instead of “faster”

- Done

Line 484: What do you mean with “direct and indirect effects of temperature”? Is this the temperature effects on productivity (resources) and physiology?

- Yes, we have clarified this also here now

Lines 487-489: It is correct but it is not very useful. Almost all mechanistic models in marine and fisheries science account for both bottom-up processes and temperature effects on the

higher trophic levels (see Heneghan et al, 2021, Lotze et al., 2019, Tittensor et al., 2021....)

- Yes, that is true. We removed this sentence.

Line 490: You should replace “plastic body growth is” by “plastic body growth was”

- Done

Line 498: You showed that MSY and Fmsy can be largely affected by changes in growth and abundance-at-size not “predicted”.

- We meant that based on what happens to body growth and size-at-age, one can “predict” (but not in a statistical sense) what happens to MSY and Fmsy.
- We have edited the sentence now “[...]largely depends on the net effect of temperature on abundance-at-size and body growth rates.” (Line 697)

Line 505: “Higher” instead of “faster”

- Done

Line 505: Which vital rates in your case? Growth is a vital rate.

- True, we now say metabolism and intake rates in particular (Line 705)

Lines 505-506: You can cite Thorson et al., 2015

- Done

Line 512: If you refer to the decrease in primary production, Lotze et al. (2019) and Heneghan et al. (2021) are not fully relevant since they mainly show the changes in consumer biomass. Kwiatkowski et al., (2020) presents the up-to-date projections in primary production (<https://doi.org/10.5194/bg-17-3439-2020>)

- We now cite Kwiatkowski et al., (2020) instead, thank you.

Lines 551-552: why the negative effects of temperature on resource carrying capacity may not reflect the actual situation in the Baltic Sea?

- We removed this part of the sentence because it was already very long. (What we meant originally was however that we simply do not know to what degree we can project resource spectra in the Baltic Sea by scaling their growth rate and carrying capacity with temperature)

Lines 552-553: It is not fully relevant to refer to the global trend in primary production. In Steinacher et al, 2010 (Kwiatkowski et al., (2020)), there are some regions where primary production is projected to increase and other regions where primary production is projected to decrease. Your study can provide some insights regarding the processes at play where productivity is projected to decrease.

- Yes, though as we see it, we first state the assumption to evaluate no or negative effects of resource carrying capacity, and how that is predicted in general, and after that how our approach can easily be modified to test scenarios with increasing carrying capacities with temperature. We do not really see a need to change this.

Line 553: Could you elaborate this: “(but see e.g. Flombaum et al. (2020))” ?

- The sentence has been edited and the text has been removed (but originally, we cited this paper to say that regions where picophytoplankton dominate can have elevated phytoplankton biomass).

Lines 555-556: Did you refer to “primary productivity”? Could you provide some details?

- The sentence has been edited and the text has been removed.

Line 559: Could you elaborate this: “ but see also Barnes et al. (2011)).”

- This has been clarified now (they did not find a significant negative effect) (Line 783)

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